

Making the exercise work means deepening knowledge in a selected topic area

The exercise can be **programming project**. The topic can be freely chosen from the topic area (secure programming) of the course.

Programming project

The result will be a working program that fits to the course topic. You need to apply secure programming principles and document these. You need to be able to complete the work in your selected environment be able to work independently. You need to return the program, source code and a report. The report includes the following items:

- General description: How the program is used and to what purpose and description of user interface specially form those parts that are not self-evident.
- Structure of the program
- Secure programming solutions: How and in which parts of the code? You should also comment the code itself. You should use a checklist, for example OWASP TOP 10 or SANS 25. Describe how issues have been solved.
- In case the program is based on earlier work, report the changes
- You should include at least manual security testing, but it is highly recommended to do more extensive testing. Report testing and also what you found and what you fixed based on testing. If you made the DevSecOps exercise it is a good idea to use the pipeline throughout the programming project.
- In case something was not yet implemented, document that as well.
- If you know there is security issue or vulnerability, document that as well.
- Suggestions for improvement, what could be implemented.

Some examples of previous years' exercise works:

<https://github.com/TeemuJal/React-App>

<https://gitlab.com/Wwuoti/automatic-vulnerability-search>

<https://github.com/mielonen/TorFisher>

<https://github.com/Supsu/turo>

<https://github.com/TeroMi/WebForum>

<https://github.com/ahola5/TurvOhj-Htyo>

<https://github.com/SamuliAP/rust-wasm-xss-demo>

<https://github.com/muhammad-aftab93/SearchableSymmetricEncryption>

<https://github.com/Tampereen-Urheilusukeltajat/Blenderi-Backend/issues/44>

<https://github.com/kelhotin/dockerfileanalyzer>

https://github.com/NEKVaaananen/COMP.SEC.300_Password_manager

https://github.com/UnoriginalOrig/PyRecommender_SePro_Project

<https://github.com/elimak1/Secure-programming-assistant/tree/main>